

Melanoma screening and public education for minority populations in the United States



Collin M. Costello, MD,^a Stephen W. Dusza, DrPH, MPH,^b Ashfaq A. Marghoob, MD,^b and Allan C. Halpern, MD^b

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The United States Preventive Services Task Force concluded that there is insufficient evidence for clinician skin cancer screening.¹ Conclusive evidence is not forthcoming, as analyses with power calculations questions the feasibility of a randomized melanoma screening trial.² It is increasingly recognized that melanoma overdiagnosis requires our attention.³ Separately, high melanoma mortality-to-incidence ratios for minority groups have garnered increased attention.

Evaluating the number needed to screen (NNS) to prevent 1 death is essential when considering screening. NNS is critical as screening aims to prevent death from biologically aggressive cancer, not increase overdiagnosis. Colon cancer screening (The United States Preventive Services Task Force grade A) has a NNS of 455.⁴ Table 1 provides estimates for melanoma NNS. As we look for opportunities to decrease the mortality-to-incidence ratios for minority groups, it is important to acknowledge that considering the NNS, screening for melanoma in minorities is impractical using our current approach and will not provide value-based care. Screening in low-mortality populations places upward pressure on overdiagnosis, strains limited dermatology resources, and spends valuable health care dollars. Other harms include false positives, psychological stress, and functional/cosmetic morbidity from nonlife-saving biopsies and excisions.

For melanoma, which can be self-detected, increasing detection pressure through public

Abbreviations used:

NNS: number needed to screen
SEER: Surveillance, Epidemiology, and End Results program

education campaigns can be tantamount to screening. If screening in minority populations is inappropriate, then awareness campaigns for this population should be scrutinized closely. Publications on minority melanoma education have increased over the last 5 years. These publications fail to address the problem that no genetic or epidemiologic factors are known to identify a high-risk subset of minority patients. Without the ability to identify and target a high-mortality minority subpopulation, efforts to increase self-detection pressure should proceed with caution.

Public education promoting the recognition of morphologically atypical pigmented lesions in minority populations is not informed by data. While early attempts are ongoing, there are no large data sets of clinical and dermoscopy images of thin invasive melanomas and their benign counterparts in this population, nor can adequate datasets be readily amassed due to the rarity of the disease. Minority populations are heterogenous, but acral melanoma comprises a majority of the lethal

From the Department of Dermatology, Mayo Clinic, Scottsdale, Arizona^a; and Dermatology Service, Department of Medicine, Memorial Sloan Kettering Cancer Center, New York, New York.^b

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Correspondence to: Collin M. Costello, MD, Department of Dermatology, Mayo Clinic, 13400 E Shea Blvd, Scottsdale, AZ 85259. E-mail: costello.collin@mayo.edu.

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Table I. Number needed to screen to prevent one death by United States race and ethnic groups

Racial/ethnic group	Melanoma incidence rate in US population per 100,000*	Melanoma prevalence per 100,000 with an average duration of survival of:		Mortality to incidence ratio†	Number needed to screen to prevent one death with 20% screening risk‡ reduction based on:	
		10 Years	20 Years		10-Year average survival	20-Year average survival
Non-Hispanic White	28.245	282.45	564.90	8.7%	20,347	10,174
Non-Hispanic American Indian/Alaska Native	7.911	79.11	158.22	13.7%	46,041	23,020
Hispanic/Latino	3.941	39.41	78.82	12.5%	101,497	50,749
Non-Hispanic Asian/Pacific Islander	1.090	10.90	21.80	22.8%	201,191	100,596
Non-Hispanic Black	0.932	9.32	18.64	30.8%	174,182	87,091

$$\text{Number Needed to Screen to Prevent One Melanoma Death} = ((1/((\text{Melanoma Incidence} \times \text{Average Duration of Survival})/100,000))/\text{Mortality to Incidence Ratio})/\text{Percent Risk Reduction by Screening})$$

*Incidence rates based on 2020 Surveillance, Epidemiology, and End Results program (SEER) data.

†SEER average melanoma SEER mortality-to-incidence rates from 2015-2019; non-Hispanic American Indian/Alaska Native mortality-to-incidence ratio is based on SEER incidence 2020 data and 2019 mortality data.

‡20% risk reduction is based on the randomized controlled trials performed for colon, breast, and lung cancer screening that showed a reduction of 18%, 20%, and 20%, respectively.

melanomas in Black, Asian, and Pacific Islander populations. A starting educational approach for public education in these populations should recognize the overall very-low mortality risk and focus on larger acral lesions (≥ 10 mm) with more than 1 color and changing color, shape, or size over months. A focus on change could improve sensitivity/specificity. Previous research in predominantly White populations suggested a clinical cutoff of 10 mm.⁵ When educating about acral lesions, the context of background ethnic pigmentation and contralateral acral pigmentation needs to be considered.

The dermatology community has made great strides in identifying and initiating research to close melanoma inequity gaps. Our campaigns against melanoma in non-Hispanic White populations have driven the earlier diagnosis of melanoma, but at the cost of increased overdiagnosis. As we look to decrease the mortality-to-incidence ratios for minority populations, we must be careful not to create a wave of unnecessary concern, biopsies, and overdiagnosis in these low-mortality populations.

Conflicts of interest

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